

# Daniel Foreman-Mackey

dfm@dfm.io, <https://dfm.io>

Member of Technical Staff, Anthropic, New York, NY

## Education

PhD 2015, Department of Physics, New York University. Advisor: Hogg

MSc 2010, Department of Physics, Queen's University, Canada. Advisor: Widrow

BSc 2008, Department of Physics, McGill University, Canada.

## Positions

Member of Technical Staff, Anthropic, 2025–present.

Research Engineer, Google DeepMind, 2024–2025.

Research Scientist, Flatiron Institute, 2022–2024.

Associate Research Scientist, Flatiron Institute, 2017–2022.

Sagan Postdoctoral Fellow, University of Washington, 2015–2017.

## Popular open-source software

**jax** — 36315 stars / 3786 forks

Composable transformations of Python+NumPy programs: differentiate, vectorize, JIT to GPU/TPU, and more [\[docs\]](#)

**tinygp** — 348 stars / 35 forks

The tiniest of Gaussian Process libraries [\[docs\]](#)

**emcee** — 1600 stars / 440 forks

The Python ensemble sampling toolkit for affine-invariant MCMC [\[docs\]](#)

**corner.py** — 576 stars / 234 forks

Make some beautiful corner plots [\[docs\]](#)

**exoplanet** — 239 stars / 57 forks

Fast & scalable MCMC for all your exoplanet needs! [\[docs\]](#)

**daft** — 687 stars / 121 forks

Render probabilistic graphical models using matplotlib [\[docs\]](#)

## Mentorship

I collaborate with and mentor many students and postdocs, often on a single project. Below is a list of the group members who I have formally mentored as part of the Flatiron Research Fellowship and Pre-doctoral Fellowship at the Center for Computational Astrophysics.

*Current postdocs:* Thavisha Dharmawardena, Jiayin Dong, Nora Eisner, Lionel Garcia, Joseph Long.

*Current students:* Quadry Chance, Soichiro Hattori.

*Former postdocs:* Megan Bedell, Trevor David, Rodrigo Luger.

*Former students:* Fran Bartolić, Eoin Farrell, Alex Gagliano, Karl Jaehnig, Gautam Nagaraj, Pa Chia Thao, Nhat Quang Hoang Tran.

## Selected invited talks & tutorials

- Open software for Astrophysics*, 2023, Invited Plenary, 241st AAS Meeting, Seattle.
- Gaussian Processes for EPRV*, 2022, Invited Tutorial, University of Oxford, UK.
- Methods for scalable probabilistic inference*, 2022, Colloquium, University of Illinois Urbana-Champaign.
- 2022, Colloquium, UC Berkeley.
- 2022, Colloquium, University of Oxford, UK.
- 2021, Invited Talk, Institute for Pure & Applied Mathematics, UCLA.
- Advanced probabilistic modeling*, 2021, Tutorial, Harley Wood Winter School of Astronomy, Australia.
- Open-source software for probabilistic data analysis in astronomy*, 2021, Seminar, Instituto de Astrofísica, Portugal.
- Gaussian processes & stellar variability*, 2021, Seminar, CARMENES Team Meeting.
- Extending JAX with custom C++ & CUDA*, 2021, Invited Talk, IRIS-HEP Topical Meeting, CERN.
- Open source software for probabilistic data analysis*, 2020, Invited Talk, OzGrav Early Career Researcher Symposium, Australia.
- The why & how of exoplanet, a domain-specific PyMC3 extension*, 2020, Contributed Talk, PyMC Con.
- A modular ecosystem for probabilistic data analysis*, 2019, Invited Talk, Open Digital Infrastructure in Astronomy conference, Kavli Institute for Theoretical Physics.
- Exoplanet population inference, a tutorial*, 2019, Invited Talk, Exostar19 conference, Kavli Institute for Theoretical Physics.
- Astronomy as a testbed for statistical method development*, 2019, Colloquium, Center for Statistics and Machine Learning, Princeton.
- Data-driven discovery in the astronomical time domain*, 2018, Colloquium, Institute for Theory and Computation, Harvard-Smithsonian Center for Astrophysics.
- 2018, Colloquium, University of California, Santa Cruz.
- 2017, Interdisciplinary Colloquium, CIERA, Northwestern University.
- A practical introduction to Gaussian Processes for astronomy*, 2017, Invited Talk, Statistical Challenges in Astrophysics, University of New South Wales, Australia.
- Long-period transiting planets & their population*, 2016, Invited talk, Exoplanets I, Davos.
- 2016, Invited talk, Statistical Challenges of Modern Astrophysics, Carnegie Mellon.
- 2016, Colloquium, Villanova.
- Scalable Gaussian processes & the search for transiting exoplanets*, 2015, Data Science at the LHC, CERN, Geneva.
- Discovery & characterization of transiting exoplanets & their population*, 2015, Colloquium, University of Washington.
- Hierarchical inference for exoplanet population inference*, 2015, IAU Symposium, Honolulu.
- Data-driven models*, 2015, Extreme precision radial velocities, Yale.
- Population inference from noisy & incomplete catalogs*, 2015, Local Group Astrostatistics, University of Michigan.
- Time series analysis, Gaussian Processes, and the search for exo-Earths*, 2014, PyData NYC

conference, New York.

*Introduction to Gaussian Processes, probabilistic graphical models, and deep learning*, 2014, Astro Hack Week, University of Washington.

*An astronomer's introduction to Gaussian processes*, 2014, Bayesian Computing for Astronomical Data Analysis (Summer school at Penn State University).

## Grants

NSF-CDS&E (PI: Agol) *Development of fast, multi-dimensional Gaussian Processes for Exoplanet discovery and beyond*, \$471,048.00, 2019–2022

NSF-AAG (PI: Agol), *Collaborative Research: Masses and architectures of (potentially habitable) exoplanet systems*, \$491,950, 2016–2018

K2 Guest Observer – Cycle 3 (PI: Penny), *Free-Floating and Bound Planet Mass Measurements with K2: Ground- and Space-Based Photometry, Event Detection and Modeling*, \$84,000, 2016–2017

K2 Guest Observer – Cycle 3 (PI: Hogg), *Ultra-precise photometry in crowded fields: A self-calibration approach*, \$100,000, 2016–2017

XSEDE (PI: Foreman-Mackey), *A systematic search for transiting exoplanets using K2*, 100,000 CPU hours, 2015–2016

## Honors

Kavli Fellow, 2015.

Sagan Postdoctoral Fellowship, 2015–2017.

James Arthur Graduate Fellowship, 2014.

Horizon Fellowship in the Natural & Physical Sciences, 2012.

Henry M. MacCracken Fellowship, 2010.

NSERC Undergraduate Summer Research Award, 2007.

## Professional service & activities

Associate Editor-in-Chief — Journal of Open Source Software, 2019–2025

Active Referee — AAS Journals, MNRAS, PASP, A&A, Journal of Statistical Software, Journal on Uncertainty Quantification, Journal of Open Source Software

Panelist — NSF, NASA, LSSTC